THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The number of seconds in 7 days is: 1

(a) 6,01,200 **(b)** 60,480 **(c)** 6,04,800 **(d)** 6,05,800

2. The value of $(125 - 5 \times (9 + 11)) \div 5 \times 3$ is: 1

(a) 15 **(b)** 25 **(c)** 13 **(d)** 17

3. $0.2 \times 0.3 \times 0.4 =$ 1

(a) 0.24 **(b)** 0.024 **(c)** 2.4 **(d)** 0.0024

4. If $a = 3$ and $b = 4$, the value of $5a^2 - 6b$ is: 1

(a) 22 **(b)** 21 **(c)** 23 **(d)** 20

5. The angle between the bisectors of the two angles of a linear pair is: 1

(a) 180° **(b)** 60° **(c)** 45° **(d)** 90°

6. The smallest 3-digit number divisible by 9 is: 1

(a) 117 **(b)** 107 **(c)** 109 **(d)** 108

7. The number of triangles with whole-number sides (in cm) and perimeter 9 cm is: 1

(a) 3 **(b)** 4 **(c)** 2 **(d)** 5

8. $\frac{3}{4} \div (\frac{3}{2} \div \frac{1}{2})$ equals: 1

(a) 1 **(b)** $\frac{1}{4}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{1}{8}$

9. The number of zeros in 25,000 × 400 is: 1

(a) 9 **(b)** 8 **(c)** 7 **(d)** 6

10. The value of 'twice the sum of 7 and 5, decreased by 9' is: 1

(a) 33 **(b)** 24 **(c)** 6 **(d)** 15

11. How many times does 0.05 go into 2.4? 1

(a) 49 **(b)** 50 **(c)** 48 **(d)** 47

12. Simplify $5(x + 2) - 3(x - 1)$: 1

(a) $2x + 13$ **(b)** $8x + 13$ **(c)** $2x - 7$ **(d)** $2x + 7$

13. Two supplementary angles are in the ratio $1 : 2$. The larger angle is: 1

(a) 130° **(b)** 140° **(c)** 110° **(d)** 120°

14. The largest 4-digit number with all digits different that is divisible by 9 is: 1

(a) 9875 **(b)** 9872 **(c)** 9874 **(d)** 9873

15. The exterior angle at the base of an isosceles triangle is 110°. The apex angle is: 1

(a) 30° **(b)** 40° **(c)** 60° **(d)** 50°

16. $\frac{2}{5}$ of a number is 18. Three times the number is: 1

(a) $67\frac{1}{2}$ **(b)** 135 **(c)** $135\frac{1}{2}$ **(d)** 270

17. 1 crore = 1,00,00,000; the number of crores in 25,00,00,000 is: 1

(a) 2.5 **(b)** 2500 **(c)** 25 **(d)** 250

18. Compare 14 × 9 and 13 × 10. The first minus the second is: 1

(a) -3 **(b)** -5 **(c)** -4 **(d)** -2

19. 0.6 of a number is 4.2. The number is: 1

(a) 8 **(b)** 7 **(c)** 9 **(d)** 6

20. The perimeter of a rectangle with sides $5x + 4$ and $3x - 1$ is: 1

(a) $30x$ **(b)** $8x + 3$ **(c)** $16x + 6$ **(d)** $4x + 10$

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. How many seconds are there in 7 days? Write it in the Indian and the international system. 2

22. Evaluate $(125 - 5 \times (9 + 11)) \div 5 \times 3$. 2

23. Find the missing number: $0.6 \times \square = 0.162$. 2

24. Simplify: $7(x - 2) - 3(x - 5)$. 2

25. Find the angle whose supplement is 3 times its complement. 2

26. A two-digit number added to its reverse gives 110 and subtracted from it gives 36. Find the number. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. A bank gives ₹2,50,000 loan to each of 48 persons. Find the total, write it in words and round it to the nearest crore. 3

28. A shop sells 24 notebooks at ₹35 each and 30 pens at ₹12 each, and gives a discount of ₹20 on every 6 notebooks. Write an expression for the total amount and find it (discount counted once for each full set of 6 notebooks). 3

29. Find the value of $2.5 \times 0.4 + 1.6 \div 0.4$. 3

30. A rectangle has length $(5x + 4)$ cm and breadth $(3x - 1)$ cm. Find its perimeter as an expression and its value for $x = 6$. 3

31. Angles on one side of a straight line at a point are 35°, 45°, 50° and a fourth angle. Find the fourth angle and name the kind of each angle. 3

OR

The number $4 * 3 * 6$ has the same digit $*$ in both blanks and is divisible by 9. Find all possible values of $*$. 3

32. In $\triangle ABC$, $\angle A = 50^\circ$. The exterior angles at B and C are $(4x + 60)^\circ$ and $(3x + 100)^\circ$. Find x and the angles B and C. 3

OR

Ravi alone can finish a job in 6 days and Sita alone in 3 days. What fraction of the job do they finish together in one day, and how many days do they take together? 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. A city has 2,50,000 houses. Each house uses 180 units of electricity per month. Find the units used by the city in one month and in a year (12 months). Write the yearly figure in words and round it to the nearest crore. 5

34. A school fete: 60 children each buy 3 tickets at ₹10 each. 40 adults buy 2 tickets at ₹25 each. The organisers spend ₹800 on prizes and ₹400 on decoration. Write one expression for their profit and evaluate it. 5

35. A tank of capacity 60 litres is filled by a pipe at 7.5 litres per minute, but a leak drains 2.5 litres per minute. Find the net gain per minute and the time to fill the empty tank. 5

36. In a pattern, figure n has $4n + 3$ tiles. Find 5
(i) the tiles in figure 15,
(ii) the total tiles in figures 1 and 2, and
(iii) how many more tiles figure 15 has than figure 14.

OR

Three angles on one side of a straight line at a point are in the ratio $1 : 2 : 3$. Find the three angles and name the kind of each. 5

37. Use reverse-and-add on 69 until a palindrome is formed. Show each step, state the number of steps and the final palindrome. 5

OR

The exterior angles of a triangle (one at each vertex) are in the ratio $3 : 4 : 5$. Find the three exterior angles, the three interior angles and classify the triangle. 5

38. A farmer's land is 240 acres. He sells $\frac{1}{4}$ of it, gives $\frac{1}{3}$ of the remainder to his son and plants $\frac{3}{4}$ of what is left after that with sugarcane. Find each area and the area left unplanted. 5

OR

Two numbers add up to 1,25,00,000 and differ by 15,00,000. Find the numbers, and write both in words. 5

END OF QUESTION PAPER — Total: 80 Marks

MASTERMIND